Yanan Bo

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Summary

- Ph.D. candidate in Computer Engineering at Clemson University (expected 2026), focused on scalable ML and distributed optimization, with a first-author PNAS publication.
- Develop communication-efficient and stable training algorithms that reduce per-iteration communication by $> 50\%$ and accelerate convergence by up to $2\times$, delivering strong time-to-accuracy gains and improved robustness.
- Build reproducible benchmarking pipelines and validate results by running training across multiple tasks and benchmarks (e.g., ImageNet-scale training).
- Aim to translate algorithmic advances into reliable, scalable ML systems by bridging theory and practice and improving efficiency, stability, and time-to-accuracy under practical constraints.

Education

Clemson University | Clemson, SC, USA — Aug 2021 – Present
Doctor of Philosophy (Ph.D.), Computer Engineering

Central South University | Changsha, China — Sep 2017 – Jun 2020
Master's Degree, Mathematics

Central South University | Changsha, China — Sep 2013 – Jun 2017
Bachelor's Degree, Mathematics and Applied Mathematics

Technical Skills

Languages: Python, C/C++, Java, Bash/Shell, MATLAB
Deep Learning: PyTorch, TensorFlow
Scientific Computing/ML: NumPy, SciPy, pandas, scikit-learn
Research Tools: Jupyter, Git, SQL, CPLEX, SPSS, LaTeX
Distributed Learning: communication compression/quantization, convergence and time-to-accuracy benchmarking, stability under heterogeneity, ablation/robustness evaluation
Training: Udacity, Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) (in progress)

Experience/Projects

Communication-Efficient Distributed Nonconvex Optimization (PNAS 2024), Clemson University

- Designed a distributed learning algorithm that reduces per-iteration **communication cost** by over **50%** through quantization, enabling scalable training under bandwidth-constrained environments.
- Demonstrated that controlled quantization noise—**leveraging quantization errors as free perturbations**—improves optimization dynamics by enabling escape from saddle points, resulting in up to **2× faster convergence** and improved training accuracy in practice.
- Built reproducible benchmarks across distributed workloads (CNN training, tensor decomposition, video background separation) with rigorous ablations and robustness checks.
- First-author publication in *Proceedings of the National Academy of Sciences* (PNAS), 2024.

Accelerated Convergence in Distributed Optimization, Clemson University

- Proposed a gradient-clipping-based distributed optimization algorithm to **stabilize gradient updates**; as an added benefit **at essentially no additional cost**, it accelerates saddle-point escape, improving iteration complexity from $\mathcal{O}(\epsilon^{-4})$ to $\mathcal{O}(\epsilon^{-3})$ (a theoretical $\mathcal{O}(1/\epsilon)$ speedup).
- Implemented and evaluated the method in Python across real-world tasks (logistic regression, background separation, and neural network training), achieving up to **4×** faster saddle escape ($\sim$500 vs. $\sim$2000 iterations) and **1.5×–2× faster convergence** to the same accuracy with consistently lower final training loss than unclipped baselines.

- Further validated on **ImageNet** CNN training, outperforming CDSGD and noisy gradient tracking with faster convergence and higher train/test accuracy, indicating improved **stability** and **generalization**.
- Conducted sensitivity and robustness checks across random seeds and algorithmic settings (e.g., noise amplitude and clipping thresholds), consistently reducing stagnation near saddle or saddle-like regions; first-author manuscript under review at *IEEE Transactions on Automatic Control* (TAC).

Decentralized Optimization under Generalized Smoothness, Clemson University

- Developed a provably convergent decentralized optimization algorithm over **directed graphs** under **generalized smoothness** conditions, integrating gradient tracking with a clipping mechanism to **improve stability** under **heterogeneous** objectives.
- Removed the widely used bounded gradient dissimilarity assumption in distributed optimization, enabling convergence guarantees under **more realistic heterogeneous data**, and proved an $\mathcal{O}(\epsilon^{-2})$ **convergence rate**, matching best-known centralized complexity under the same smoothness regime.
- Validated on benchmark workloads, including decentralized ResNet-18 training on CIFAR-10, achieving **1.5× faster epoch-to-accuracy** than the fastest baseline (same test accuracy in 100 vs. 150 epochs; same training accuracy in 300 vs. 450 epochs) with improved stability.
- First-author manuscript under review at *IEEE Transactions on Automatic Control* (TAC).

Data-Driven Design and Optimization of Reverse Logistics Networks, Central South University

- Built a data-driven (nonlinear) mixed-integer optimization model for large-scale WEEE reverse logistics networks, integrating recycling prices, transportation costs, processing capacity, and subsidies for facility location and flow allocation.
- Modeled consumer participation via endogenous recycling prices and solved large instances with branch-and-bound–based methods; validated performance across multiple network configurations.
- Conducted a real-world case study in Central Hunan, China with sensitivity analysis (subsidy, transportation modes, environmental awareness), achieving a **12.5%** profit-to-cost margin; published first-author in *Journal of Advanced Transportation* (2019).

Teaching Experience

Co-instructor, Clemson University
Courses: Introduction to Digital Signal Processing
- Guided students in MATLAB-based implementation of digital signal processing techniques, reinforcing theoretical concepts through hands-on coding and problem-solving support.

Publications

[1] **Y. Bo**, Y. Wang. Quantization avoids saddle points in distributed optimization[J]. *Proceedings of the National Academy of Sciences (PNAS)*, 121(17), e2319625121, 2024. DOI: 10.1073/pnas.2319625121.
[2] **Y. Bo**, Y. Wang, Z Wan. Optimizing the WEEE recovery network associated with environmental protection awareness and government subsidy by nonlinear mixed integer programming[J]. *Journal of Advanced Transportation*, 2019(1): 9858670.
[3] **Y. Bo**, Y. Wang. Provably convergent decentralized optimization over directed graphs under generalized smoothness. *arXiv preprint* arXiv: 2601.03566, 2026. (Submitted to *IEEE Transactions on Automatic Control (TAC)*, under review.)
[4] **Y. Bo**, Y. Wang. Gradient clipping accelerates saddle avoidance in distributed optimization[J]. *IEEE Transactions on Automatic Control (TAC)*, under review.

Honors and Awards

- National Scholarship (top graduate student honor, China), 2020
- Outstanding Graduate, Central South University (university-level honor), 2020

Media & Outreach

- Featured in Clemson University News for research on distributed optimization, published in the *Proceedings of the National Academy of Sciences* (PNAS), 2024.